UMKC Impact & Translation Series

Assessing the Market for your Innovation

Helping you move your work beyond the university

An innovation needs to come out of the lab to make an impact. For universities and research institutions, transforming a groundbreaking idea into real-world use requires careful planning and strategic evaluation.

This whitepaper aims to equip researchers and innovators at UMKC with a framework to assess the market potential of their innovations. Drawing from expert insights, we provide a dynamic guide for market analysis, technology evaluation, and strategic planning.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: how to assess the market for your innovation.

Understanding Market Analysis

"Market analysis" is the process of understanding the environment in which an innovation will exist. How you define the market provides a compass to guide your innovation through the competitive landscape and towards real-world use. It combines quantitative data (market size, revenue projections) with qualitative insights (consumer values, buying motives) to forecast the potential impact of your product or service.

A thorough market analysis will identify the following elements:

1. **Target Industry:** Dive deep into industry trends, market size, and the regulatory landscape to understand where your innovation fits.
2. **Competitive Landscape:** Identify key players, analyze their strengths and weaknesses, and uncover opportunities to differentiate your product.
3. **Identification of Market Gaps:** Look for unmet needs and underserved segments that your innovation can address.
4. **Definition of Target Market:** Segment your potential customers based on demographics, psychographics, and behavioral patterns.
5. **Barriers to Entry:** Assess challenges such as startup costs, regulatory requirements, and competitive pressures.
6. **Sales Forecast:** Project future sales to inform your business strategy and attract potential investors.

Conducting Market Research

Market research helps test whether your work connects to a real need outside the university. It is your window into the minds of potential customers. By gathering and analyzing data, you can validate your product's viability and refine your go-to-market strategy.

Market research fits into two categories:

- **Primary Research:** Data collected directly through surveys, focus groups, and interviews to gain first-hand insights.
- **Secondary Research:** Data collected from industry reports, academic studies, and government statistics to inform your analysis.

Primary Research

Primary research is used to learn about the specific pain points and needs that relate to values your invention can address. **Surveys** give statistical backing, **focus groups** provide collective insights, and **interviews** offer deep, individualized perspectives. These tools provide quantifiable and qualitative measures for how your invention will solve problems, giving you data to define the commercial potential of your invention.

Developing a Survey

Surveys are a cost-effective and easy-to-implement option to create quantifiable data from a broad audience of potential users.

To develop an effective survey, consider these tips:

1. **Define Objectives Clearly** – Identify what specific information you need (e.g., customer pain points, willingness to pay, competitive landscape).
2. **Keep It Concise** – Shorter surveys (5–10 minutes) have higher completion rates.
3. **Use a Mix of Question Types:**
    a. Closed-ended: Multiple choice, Likert scales (1-5 or 1-7), ranking. (Author's Note: I recommend an even number of choices on Likert scales. Instead of providing a middle option that is the equivalent of "no choice," an even number of responses (ie, 1-4 or 1-6) forces respondents to choose a positive or negative response, better relating to your evaluation.)
    b. Open-ended: Allow respondents to provide insights in their own words.
4. **Avoid Leading Questions** – Ensure neutrality in your questions (e.g., "What challenges do you face?" instead of "Do you struggle with X problem?"). Poorly worded or leading questions can lead to skewed results for your survey.
5. **Segment Your Audience** – Identify ideal respondents based on industry, job role, and experience level. Understand the roles these individuals will play in purchasing, using, or benefitting from your invention.

6. **Pilot Test the Survey** – Run a small test with a sample group before a full launch to identify issues.
7. **Choose the Right Distribution Method** – Email lists, industry forums, LinkedIn groups, or professional associations. UMKC has several survey tools at our disposal; check with your department, IT lead, or contact the University libraries to learn more.
8. **Offer an Incentive (If Possible)** – Small incentives like gift cards or entry into a prize draw improve response rates. It is NOT necessary for them to be expensive; people will complete surveys for a chance to win a $5 gift card. Make it something relevant and valuable for the audience you're targeting, and you'll increase the likelihood of hitting response targets.

Leading a Focus Group

Focus groups provide qualitative insights and deeper understanding of potential customer perspectives. The goal is to get potential customers to share opinions that could influence their decision to adopt your invention.

To ensure a successful session:

1. **Select the Right Participants** – Aim for 6-10 individuals who represent your target market (diversity in roles and perspectives is beneficial). Seek individuals that could be decision makers, end users, or beneficiaries of your invention and learn what is important to them.
2. **Define Clear Goals** – Outline specific discussion topics (e.g., usability of your invention, market needs, pricing expectations).
3. **Prepare a Discussion Guide** – Structure the conversation with open-ended questions to encourage dialogue.
4. **Use a Skilled Moderator** – Someone experienced in leading discussions should guide the group to prevent bias and dominance by a single voice.
5. **Encourage Honest Feedback** – Create a comfortable environment where participants feel safe sharing honest opinions.
6. **Record (With Permission)** – Video or audio recordings help ensure accurate analysis of responses.
7. **Synthesize Key Takeaways** – Identify recurring themes and actionable insights rather than individual comments.
8. **Run Multiple Groups (if feasible)** – Holding 2-3 groups with different segments of the market can provide more robust validation and different insights based on the group's composition.

Customer Discovery through Interviews

Direct interviews with potential customers offer in-depth qualitative feedback for your invention's potential impact.

Here's how to conduct effective customer interviews:

1. **Identify the Right Interviewees** – Choose people who represent decision-makers, end-users, or influencers in the buying process.
2. **Structure the Interview but Stay Flexible** – Prepare a list of key questions but allow room for organic discussion.
3. **Start Broad, Then Get Specific** – Begin with general industry questions before asking about their specific challenges and how your invention might fit.
4. **Ask Open-Ended Questions** – Encourage elaboration (e.g., "Can you describe a recent challenge you faced with [problem your invention solves]?").
5. **Actively Listen and Probe Deeper** – Follow up on interesting responses with "Can you tell me more about that?" or "Why do you think that happens?"
6. **Avoid Selling** – The goal is to gather insights, not pitch your solution.
7. **Take Detailed Notes or Record (with consent)** – This ensures accuracy and lets you focus on the conversation.
8. **Analyze Themes and Patterns –** After multiple interviews, look for consistent pain points, unmet needs, and potential market opportunities.

Secondary Research

Secondary research leverages existing data to analyze market potential, trends, and competitive landscapes.

Defining TAM, SAM and SOM

Understanding market size is essential for assessing commercial viability. These three metrics provide a structured approach:

1. **Total Addressable Market (TAM)** – The overall revenue opportunity if 100% of the market adopted your innovation.
    - Example: If your invention is a novel wound dressing, TAM would include all global wound care product sales.
2. **Serviceable Available Market (SAM)** – The portion of the TAM your company could realistically serve based on geographic, regulatory, or industry constraints.
    - Example: If your wound dressing is only available in the U.S., your SAM is the U.S. wound care market.
3. **Serviceable Obtainable Market (SOM)** – The fraction of SAM your company can capture, based on competitive positioning and go-to-market strategy.
    - Example: If your startup can realistically gain 5% of the U.S. market within five years, that's your SOM.

These metrics can be defined using several methods:

1. **Use industry reports** – Sources like IBISWorld, Gartner, Frost & Sullivan, and Statista provide market size estimates.

2. **Analyze government and trade association data** – The Census Bureau, FDA, and industry organizations publish relevant market statistics.
3. **Use competitive benchmarks** – Review financial reports from established companies in your space to estimate market share.

Using AI Tools like ChatGPT

AI tools like ChatGPT can streamline secondary research by summarizing vast amounts of data, generating insights, and identifying trends, but there are limitations. AI-generated information requires fact checking against reliable sources, as the data sources may be outdated, incomplete, or skewed.

Some methods for using AI tools in market analysis include:

1. **Industry Trend Analysis** - "What are the latest trends in [industry]?" AI can summarize reports, patents, and news to identify emerging technologies.
2. **Competitor Analysis** - "Who are the key competitors in [market]?" AI can help analyze company strategies, funding, and innovations.
3. **Customer Pain Points** - "What are common challenges faced by customers in [industry]?" AI can synthesize customer feedback from reviews, forums, and case studies.
4. **Generating Hypotheses for Testing** - AI can help structure survey questions, interview scripts, and market hypotheses to validate with primary research.

Keep in mind, some AI tools use information you put into their system to train models of data. Before you enter anything that could be considered proprietary, confirm that your AI tools meet required security standards for your department.

Using the Library

UMKC's libraries provide free access to premium databases and expert resources. Librarians are trained to know what resources are available and guide you to the proper tools for your work.

When using the library, consider the following resources:

1. **Industry & Market Reports -** Access reports from IBISWorld, Statista, MarketLine, Frost & Sullivan, and Gartner. Use university subscriptions to analyze market demand and trends.
2. **Patent & Intellectual Property Research -** Use Google Patents, USPTO, WIPO, and university patent databases to check for similar technologies. Review prior art to assess potential barriers to commercialization.
3. **Academic & Technical Publications -** Databases like PubMed, IEEE Xplore, ScienceDirect, and Web of Science help analyze technological developments.
4. **Business & Economic Data -** Use government databases (e.g., Census Bureau, Bureau of Labor Statistics) for industry size and employment trends. Analyze trade association reports for sector-specific insights.

5. **Librarian Assistance -** University business librarians can guide researchers to the most relevant datasets.

Understanding Competition

A critical component of market analysis is understanding competition. Even if an invention is groundbreaking, alternative solutions, substitutes, or indirect competitors exist.

What's the worst answer you can give? "My solution doesn't have competition." - it signals a lack of market awareness and a failure to understand customer alternatives.

Understanding competition is part of understanding whether your work is grounded in real-world conditions. Investors, industry partners, and commercialization teams expect a well-researched competitive landscape to assess an invention's viability, and that includes the customer's ability to continue life without your invention.

A competitive matrix allows you to demonstrate your research and understanding of how your invention fits in the market.

Developing a Competitive Matrix

A competitive matrix is a structured tool to compare key market players and position an invention effectively. It helps identify gaps, strengths, and differentiators.

1. **Identify Competitors**
    - Direct competitors - same solution, same market.
    - Indirect competitors - alternative solutions that solve the same problem (including no solution).
    - Future competitors - companies developing similar technologies.
2. **Select Comparison Factors**
    - Features & Functionality – What does each competitor offer?
    - Pricing & Business Model – How do they charge? What are margins?
    - Market Share & Customer Base – Who are their primary customers?
    - Intellectual Property & Innovation – Do they have patents or proprietary technology?
    - Regulatory Approvals – Are there legal barriers for market entry?
    - Customer Satisfaction & Reviews – What are common complaints or praises?
3. **Build the Matrix -** Use a simple table to compare competitors across the selected factors:

Feature/Factor	**Competitor A**	**Competitor B**	**Your Invention**
Price	$$$	$$	$$
Ease of Use	Medium	High	High
Key Differentiator	AI-driven	Manual	Real-time data

Patent Protection	Yes	No	Yes

4. **Analyze the Gaps**
    - Find **underserved areas** in the market where your invention excels.
    - Identify **barriers to entry** competitors have already overcome.
    - Look for **differentiation opportunities** that set your invention apart.

Defining your Market Position

After building a competitive matrix, the next step is positioning. Rather than just being "better," an invention should focus on value creation.

1. **Avoid Competing on Price Alone** - If an innovation is unique, **pricing too low** may signal low value. If competitors have lower-cost solutions, emphasize **ROI and efficiency** rather than just cost.
2. **Emphasize Unique Strengths -** Identify 1-2 **key differentiators** and focus messaging around them. Examples:
    - **Faster results** (e.g., diagnostic tools that reduce waiting time).
    - **Improved accuracy** (e.g., AI-driven analytics with 98% precision).
    - **More user-friendly** (e.g., a no-code software alternative to complex solutions).
3. **Leverage Customer Pain Points -** Study competitor **negative reviews** to find pain points customers want solved. Position the invention as an improvement in those areas.
4. **Consider Go-To-Market Advantages -** Speed to market, partnerships, or regulatory fast-tracking can be strategic advantages. Example: If competitors require expensive FDA approvals, a non-regulated alternative could be a major selling point.

A strong competitive analysis proves market understanding and guides positioning strategy. The goal is not just to show where an invention fits, but also to demonstrate why it offers a compelling advantage over existing solutions.

Building Barriers to Entry

Barriers to entry are obstacles that make it difficult for new entrants to enter a market. Understanding these barriers helps inventors assess the challenges they may face in commercialization and develop strategies to overcome them. Barriers can be structural (inherent to the market) or strategic (imposed by competitors or regulations).

This section outlines key barriers to entry, how they impact commercialization, and strategies for overcoming them.

Key Barriers to Entry

1. **Regulatory & Legal Barriers** - Many industries require compliance with government regulations, certifications, and safety standards before a product can be sold.
    - Examples: FDA approval for medical devices, environmental and safety regulations for industrial products, data privacy laws (e.g., GDPR, HIPAA) for software solutions.
    - Impact: Lengthy approval processes can delay commercialization and increase costs.
2. **Intellectual Property (IP) & Patent Restrictions** - Existing patents can block market entry if a similar technology is already protected. Conversely, a lack of IP protection makes an innovation vulnerable to imitation.
    - Examples: A competitor holds a broad patent that prevents similar innovations, your invention lacks patent protection, making it easy to copy.
    - Impact: Without a strong IP strategy, investors may see high legal risks.
3. **High Capital Requirements** - Some industries require significant investment in R&D, manufacturing, equipment, or marketing before a product becomes commercially viable.
    - Examples: Biotech startups needing millions for clinical trials, semiconductor companies requiring expensive fabrication facilities.
    - Impact: Lack of funding can slow or halt commercialization.
4. **Market Dominance by Established Competitors** - Well-established players often have strong brand recognition, customer loyalty, and economies of scale that make it difficult for new entrants to compete.
    - Examples: A startup creating a new search engine struggles against Google, a small medical device company competes with Johnson & Johnson.
    - Impact: Competitors can undercut pricing, block distribution channels, or use legal tactics to deter new entrants.
5. **Customer Adoption & Trust** - Convincing customers to switch to a new, unproven solution can be a major hurdle.
    - Examples: Hospitals hesitant to adopt a new medical device due to liability concerns, B2B clients unwilling to change existing workflows for a new software tool.
    - Impact: Slow adoption rates can delay revenue generation.
6. **Distribution & Supply Chain Barriers** - Even if an innovation is ready for market, limited access to distribution channels, supply chains, or retailers can restrict sales.
    - Examples: Consumer goods startups struggling to secure shelf space at major retailers, a food-tech company unable to meet supply chain requirements for nationwide distribution.
    - Impact: Limited scalability and growth potential without the right distribution partners.

Identifying barriers to entry helps inventors prepare for challenges before entering the market.

Defining your Target Market

A well-defined target market ensures an invention aligns with the right customer needs, increasing the likelihood of successful commercialization. Identifying the right market segment involves analyzing who the customers are, what motivates them, and how they behave.

To effectively define a target market, segmentation should be approached from three perspectives:

- **Demographics** (Who are they?)
- **Psychographics** (Why do they buy?)
- **Behavioral Patterns** (How do they engage with products?)

Segmenting Customers by Demographics

Demographic segmentation helps categorize potential customers based on measurable factors. This is useful for B2C (business-to-consumer) and B2B (business-to-business) markets.

For B2C (Individual Customers):

1. **Age** – Are they millennials, Gen X, or baby boomers? Each age group has different purchasing behaviors.
2. **Gender** – Some innovations cater to specific genders (e.g., women's healthcare devices).
3. **Income Level** – Does the invention serve a premium or budget-conscious audience?
4. **Education Level** – Highly technical products may appeal to an educated customer base.
5. **Location** – Urban vs. rural, domestic vs. international markets.
6. **Occupation** – Professionals, students, or retirees may have different needs.

For B2B (Organizations and Companies):

1. **Industry** – Healthcare, manufacturing, education, etc.
2. **Company Size** – Small startups vs. large enterprises.
3. **Revenue & Budget** – Can they afford premium solutions, or do they need cost-effective options?
4. **Geographic Reach** – Local businesses vs. global corporations.

Example: A smart medical diagnostic tool might target hospitals (B2B) in metropolitan areas that serve high-income patients (B2C) in their 50s and 60s who are proactive about health monitoring.

Segmenting Customers by Psychographics

Psychographic segmentation helps understand customer motivations, values, and lifestyle choices.

1. **Pain Points & Needs** – What problems do they experience that the invention solves?
2. **Values & Beliefs** – Are they driven by sustainability, innovation, or cost-effectiveness?
3. **Lifestyle & Interests** – Do they prioritize convenience, health, or technology?

4. **Buying Motivations** – Are they early adopters of technology, price-sensitive, or brand-loyal?
5. **Risk Tolerance** – Are they comfortable trying new solutions, or do they prefer proven products?

Example: A biodegradable packaging alternative would appeal to eco-conscious consumers and sustainability-driven companies that prioritize reducing their carbon footprint.

Segmenting Customers by Behavioral Patterns

Behavioral segmentation categorizes customers based on how they interact with products and make purchasing decisions.

1. **Usage Frequency** – Are they daily users, occasional users, or one-time buyers?
2. **Purchase Behavior** – Do they research extensively before buying, or make impulse decisions?
3. **Brand Loyalty** – Are they brand-switchers or committed to long-term relationships?
4. **Buying Channel Preference** – Do they prefer online shopping, in-store purchases, or B2B negotiations?
5. **Price Sensitivity** – Are they bargain hunters or willing to pay a premium for high-quality solutions?

Guiding Question:

If this work were to create impact beyond the university, who would benefit, what problem would it solve, and what would need to be true for it to be adopted or used?

Example: A smart home automation device might target tech-savvy homeowners who are early adopters of new gadgets, purchase online via e-commerce platforms, and value convenience and security over price.

Defining a target market is more than just identifying potential buyers—it's about understanding their needs, behaviors, and decision-making processes. The strongest commercialization strategies combine all three segmentation approaches to refine messaging, pricing, and distribution channels.

Using Market Analysis to Develop Sales Forecasts

A well-informed sales forecast is a key component of commercialization planning. It allows innovators to estimate future revenue, plan production and distribution, and attract investors by demonstrating potential market success. A strong sales forecast leverages insights from previous market analysis steps, including target market segmentation, competition analysis, and barriers to entry.

To build a sales forecast, follow these structured steps:

1. **Define Your Assumptions** - Before projecting sales, define the key assumptions that influence your estimates:
    - Market Penetration Rate – Based on your TAM, SAM, and SOM, estimate what percentage of your obtainable market you can realistically capture.
    - Customer Acquisition Rate – How quickly can you convert leads into paying customers?
    - Pricing Strategy – What is your expected price point per unit or service?
    - Sales Cycle – How long does it take for a customer to go from awareness to purchase? (e.g., enterprise sales may take months, while consumer products may sell instantly).
    - Seasonality & Trends – Are there expected fluctuations based on industry trends or economic factors?
2. **Estimate the Number of Potential Customers** - Using demographic, psychographic, and behavioral segmentation, determine the number of potential customers in your SOM (Serviceable Obtainable Market).
3. **Develop a Sales Projection Model** - Create a realistic sales projection by estimating the number of units sold or services delivered over a specific period (e.g., monthly, quarterly, or annually).

    Example of a first-year sales projection for a B2C product:

Month	Estimated Customers	Conversion Rate	Units Sold	Revenue ($)
Jan	5,000	2%	100	$10,000
Feb	6,000	3%	180	$18,000
Mar	7,500	4%	300	$30,000

    For B2B models, forecasts should consider:

    - Number of target companies
    - Adoption rates for new technology
    - Contract values and renewal cycles

4. **Factor in Barriers to Entry & Competition** - Your sales forecast should account for competitive threats and industry barriers:
    - Regulatory Delays – If approval is needed (e.g., FDA, patents), factor in launch delays.
    - Competitor Response – If competitors lower prices or improve offerings, market share growth may be slower.
    - Customer Adoption Curve – If the product requires behavioral change, expect slower early adoption.
5. **Validate with Sensitivity Analysis** – Consider creating three forecast scenarios:
    - Optimistic: Rapid adoption, high growth, minimal barriers.
    - Realistic: Expected adoption rate, moderate growth, standard challenges.
    - Pessimistic: Delays in adoption, regulatory issues, strong competition.

This helps prepare for risk and gives investors confidence in the forecast's robustness.

Note on Investor Expectations and Projections

Investors will **always** discount your projections.

When presenting a sales forecast, avoid labeling your estimates as "conservative" or "modest"—investors assume projections are optimistic and will automatically adjust them downward based on risk factors, competitive pressures, and market uncertainties.

Instead, focus on:

- Clearly defining the assumptions behind your forecast.
- Using data-driven justifications for your market size and growth rates.
- Presenting multiple scenarios (optimistic, realistic, and pessimistic) to demonstrate awareness of market dynamics.

Rather than underselling potential growth, provide a realistic yet compelling projection backed by strong market research. Investors are more likely to engage when they see a well-reasoned, data-supported outlook rather than an artificially low "conservative" estimate.

Summary

Moving an innovation beyond the university requires more than just a great idea - it requires strategic market analysis, competitive positioning, and financial forecasting. This whitepaper provided a structured approach to help UMKC faculty researchers and graduate students assess the commercial potential of their innovations.

Key takeaways from this framework include:

- **Market Research Foundations** – Conduct primary research (surveys, focus groups, interviews) and leverage secondary research (industry reports, AI tools, library resources).
- **Competitive Analysis** – Develop a competitive matrix to understand industry players and position an innovation effectively. Never assume "no competition" exists—alternative solutions and substitutes always influence the market.
- **Understanding Barriers to Entry** – Identify regulatory, financial, and operational challenges that could impact commercialization and develop mitigation strategies.
- **Target Market Definition** – Segment customers based on demographics, psychographics, and behavioral patterns to refine marketing and sales efforts.
- **Sales Forecasting** – Use TAM, SAM, and SOM data to estimate revenue, account for industry challenges, and build realistic projections for commercialization success.

By following this structured approach, innovators can validate their ideas, attract funding, and build a solid commercialization strategy that maximizes the impact of their research.

Resources & Contacts

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